



The Islamia University of Bahawalpur

Assignment # 1 | Theory of Computation

Regular Expression to recognize an Email Address and DFA of the Regular Expression

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REGULAR EXPRESSION TO RECOGNIZE AN EMAIL ID

SYNTAX:

`[username]@[domain].[extension]`

REGEX:

`[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}`

EXPLANATION:

- `[a-zA-Z0-9._%+-]+`

a-z	lowercase letters
A-Z	uppercase letters
0-9	digits
._%+-	special allowed characters
+	One or more chracters

- `@[a-zA-Z0-9.-]+\.`

@	This is the mandatory separator
a-zA-Z	Uppercase and lower letters
0-9	digits
.-	special allowed characters
+	One or more chracters
\.	a literal dot

- `[a-zA-Z]{2,}`

[a-zA-Z]	Only letters allowed
{2,}	at least 2 letters

SUMMARY

Part	Meaning
[...]+	username
@	separator
[...]+	domain
\.	dot
{2,}	extension (min 2 letters)

DFA

$$M = (Q, \Sigma, \delta, q_0, F)$$

ALPHABET (Σ)

We simplify symbols into categories (very important for TOC):

$$\Sigma = \{a, d, s, @, ., -\}$$

Where:

a	any letter [a-zA-Z]
d	any digit [0-9]
s	special symbols {_, %, +}
@	at symbol
.	dot
-	hyphen

STATES (Q)

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_d\}$$

q₀	Start
q₁	Username
q₂	After @
q₃	Domain
q₄	Dot found
q₅	Extension (Final)
q_d	Dead state

DIAGRAM

